

Exercise Sheet 9 – Solutions

Linear Algebra

Exercise 35

Problem. Linear transformation $T: \mathbb{R}^5 \rightarrow \mathbb{R}^3$,

$$T(a, b, c, d, e) = (b + c + 4d + 2e, a + c + 3d + e, 4a - 3b + c - 2e).$$

- (a) Basis and dimension of $\text{Ker}(T)$.
- (b) Basis and dimension of $\text{Im}(T)$.
- (c) General form of $v \in \mathbb{R}^5$ with $T(v) = (1, 2, 5)$.
- (d) Is T injective, surjective, bijective? (Should follow *from* (a), (b) without new computation.)
- (e) Is T invertible? If yes: explicit formula for T^{-1} .

What do the terms mean?

- **Linear transformation** $T: V \rightarrow W$: function that respects addition and scalar multiplication.
- **Kernel** $\text{Ker}(T) = \{v \in V : T(v) = 0\}$. All vectors that are mapped to the zero vector. It is a subspace of V .
- **Image** $\text{Im}(T) = \{T(v) : v \in V\}$ = all reachable values. Subspace of W .
- **Representation matrix** A of T : columns are the images of the standard basis vectors $T(e_1), T(e_2), \dots$. Then $T(v) = Av$.
- **Injective** $\iff \text{Ker}(T) = \{0\} \iff \dim \text{Ker}(T) = 0$.
- **Surjective** $\iff \text{Im}(T) = W \iff \dim \text{Im}(T) = \dim W$.
- **Bijective** $\iff$ injective *and* surjective $\iff T$ invertible.
- **Rank-Nullity Theorem**: $\dim \text{Ker}(T) + \dim \text{Im}(T) = \dim V$ (domain).

What needs to be done?

1. Set up representation matrix A (columns = $T(e_i)$).
2. Do Gaussian elimination once – with this we can read off (a), (b), (c) directly.
3. **(a)** Kernel: solve homogeneous system $Av = 0$. Free variables $\rightarrow$ basis vectors.
4. **(b)** Image: mark pivot columns in the row echelon form $\rightarrow$ corresponding *original columns* of A form the basis.
5. **(c)** Particular solution of $Av = (1, 2, 5)$ + general kernel solution.
6. **(d)** Conclude directly from $\dim \text{Ker}$ and $\dim \text{Im}$.
7. **(e)** Bijectivity $\iff$ invertible.

Step 1: Set up representation matrix

Images of the standard basis vectors:

$$\begin{aligned}T(e_1) &= T(1, 0, 0, 0, 0) = (0, 1, 4) \\T(e_2) &= T(0, 1, 0, 0, 0) = (1, 0, -3) \\T(e_3) &= T(0, 0, 1, 0, 0) = (1, 1, 1) \\T(e_4) &= T(0, 0, 0, 1, 0) = (4, 3, 0) \\T(e_5) &= T(0, 0, 0, 0, 1) = (2, 1, -2)\end{aligned}$$

These five vectors are the **columns** of A :

$$A = \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 3 & 1 \\ 4 & -3 & 1 & 0 & -2 \end{pmatrix}.$$

Step 2: Row echelon form of A (once for all subproblems)

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 3 & 1 \\ 4 & -3 & 1 & 0 & -2 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & 0 & 1 & 3 & 1 \\ 0 & 1 & 1 & 4 & 2 \\ 4 & -3 & 1 & 0 & -2 \end{pmatrix} \xrightarrow{R_3 \rightarrow R_3 - 4R_1} \begin{pmatrix} 1 & 0 & 1 & 3 & 1 \\ 0 & 1 & 1 & 4 & 2 \\ 0 & -3 & -3 & -12 & -6 \end{pmatrix} \\ \xrightarrow{R_3 \rightarrow R_3 + 3R_2} \begin{pmatrix} \boxed{1} & 0 & 1 & 3 & 1 \\ 0 & \boxed{1} & 1 & 4 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Pivots in **column 1 and 2** (corresponding to variables a, b). Free variables: c, d, e .

(a) Kernel of T

Solve $Av = 0$. Row echelon form $\Rightarrow$ equations:

$$\begin{aligned}\text{Row 1: } a + c + 3d + e &= 0 \Rightarrow a = -c - 3d - e \\ \text{Row 2: } b + c + 4d + 2e &= 0 \Rightarrow b = -c - 4d - 2e\end{aligned}$$

Rename free variables: $c = s, d = t, e = u$.

$$(a, b, c, d, e) = (-s - 3t - u, -s - 4t - 2u, s, t, u).$$

Split according to s, t, u :

$$\begin{pmatrix} a \\ b \\ c \\ d \\ e \end{pmatrix} = s \begin{pmatrix} -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -3 \\ -4 \\ 0 \\ 1 \\ 0 \end{pmatrix} + u \begin{pmatrix} -1 \\ -2 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

Basis of $\text{Ker}(T)$: $\{(-1, -1, 1, 0, 0), (-3, -4, 0, 1, 0), (-1, -2, 0, 0, 1)\}$. $\dim \text{Ker}(T) = 3$.
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(b) Image of T

$\text{Im}(T) = \text{span}\{T(e_1), T(e_2), T(e_3), T(e_4), T(e_5)\}$, i.e. the column space of A .

Mark pivot columns of the row echelon form $\rightarrow$ corresponding original columns form basis. Pivots in columns 1, 2 $\Rightarrow$ basis columns are the *first two* columns of the *original matrix* A :

Column 1 of $A = T(e_1) = (0, 1, 4)$, Column 2 of $A = T(e_2) = (1, 0, -3)$.

$\text{Basis of } \text{Im}(T): \{(0, 1, 4), (1, 0, -3)\}.$ $\dim \text{Im}(T) = 2.$

Check Rank-Nullity Theorem: $3 + 2 = 5 = \dim \mathbb{R}^5$. ✓

(c) General v with $T(v) = (1, 2, 5)$

Augmented matrix with right-hand side $(1, 2, 5)^\top$, same row operations as in Step 2:

$$\left(\begin{array}{ccccc|c} 0 & 1 & 1 & 4 & 2 & 1 \\ 1 & 0 & 1 & 3 & 1 & 2 \\ 4 & -3 & 1 & 0 & -2 & 5 \end{array} \right) \rightarrow \left(\begin{array}{ccccc|c} \boxed{1} & 0 & 1 & 3 & 1 & 2 \\ 0 & \boxed{1} & 1 & 4 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

No contradiction row $\Rightarrow$ consistent.

Particular solution (set $c = d = e = 0$): $a = 2$, $b = 1$.

$$v_0 = (2, 1, 0, 0, 0).$$

Verification: $T(2, 1, 0, 0, 0) = (1 + 0 + 0 + 0, 2 + 0 + 0 + 0, 8 - 3 + 0 - 0) = (1, 2, 5)$. ✓

General solution = particular solution + kernel:

$v = (2, 1, 0, 0, 0) + s(-1, -1, 1, 0, 0) + t(-3, -4, 0, 1, 0) + u(-1, -2, 0, 0, 1),$ $s, t, u \in \mathbb{R}.$
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(d) Injective / Surjective / Bijective

From (a): $\dim \text{Ker}(T) = 3 \neq 0 \Rightarrow T$ is **not injective**.

From (b): $\dim \text{Im}(T) = 2 \neq 3 = \dim \mathbb{R}^3 \Rightarrow T$ is **not surjective**.

Since bijective = injective *and* surjective, T is also **not bijective**.

(e) Invertible?

T is invertible $\iff T$ is bijective. From (d): not bijective $\Rightarrow T$ is **not invertible**.

(Alternative justification: for $T: \mathbb{R}^5 \rightarrow \mathbb{R}^3$ the domain and codomain cannot have the same dimension, so T can never be bijective.)

Exercise 36

Problem. $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ with $T(a, b, c) = (3b + c, 2a + b, a)$.

(a) Is T invertible? If yes: explicit formula for T^{-1} .

(b) From (a): determine kernel and image (simplest descriptions).

(c) From (a): injective / surjective / bijective?

What do the terms mean?

- **Inverse:** T^{-1} “undoes” T . For $(x, y, z) = T(a, b, c)$, T^{-1} recovers the original vector (a, b, c) from (x, y, z) .
- **Search strategy** for T^{-1} : solve the equations $T(a, b, c) = (x, y, z)$ for a, b, c .

(a) Invertibility and T^{-1}

Step 1: Set up equations. $(x, y, z) = T(a, b, c)$ means:

$$3b + c = x$$

$$2a + b = y$$

$$a = z$$

Step 2: Solve from bottom to top.

(a) a from equation 3: $a = z$.

(b) b from equation 2: $b = y - 2a = y - 2z$.

(c) c from equation 1: $c = x - 3b = x - 3(y - 2z) = x - 3y + 6z$.

Unique solution for every $(x, y, z) \Rightarrow T$ is invertible.

$$T^{-1}(x, y, z) = (z, y - 2z, x - 3y + 6z).$$

Verification: $T(T^{-1}(x, y, z)) = T(z, y - 2z, x - 3y + 6z)$:

- 1st comp.: $3(y - 2z) + (x - 3y + 6z) = 3y - 6z + x - 3y + 6z = x$ ✓
- 2nd comp.: $2z + (y - 2z) = y$ ✓
- 3rd comp.: z ✓

(b) Kernel and Image

T is invertible $\Rightarrow T$ is bijective $\Rightarrow$:

$$\text{Ker}(T) = \{(0, 0, 0)\} \quad (\text{only the zero vector})$$

$$\text{Im}(T) = \mathbb{R}^3 \quad (\text{the entire codomain})$$

(c) Injective / Surjective / Bijective

T invertible $\iff T$ bijective $\iff T$ injective and surjective.

T is **injective, surjective, bijective.**

Exercise 37

Problem. $T: \mathbb{R}^{2 \times 2} \rightarrow \mathbb{R}^{2 \times 2}$ with $T(X) = M \cdot X$, where $M = \begin{pmatrix} 1 & 2 \\ 4 & 8 \end{pmatrix}$.

- Basis and dimension of $\text{Ker}(T)$.
- Basis and dimension of $\text{Im}(T)$.

What do the terms mean?

- The vector space here is $\mathbb{R}^{2 \times 2}$ (all 2×2 matrices). $\dim \mathbb{R}^{2 \times 2} = 4$.
- **Observation:** In $M = \begin{pmatrix} 1 & 2 \\ 4 & 8 \end{pmatrix}$, row 2 = 4 · row 1, and column 2 = 2 · column 1. So M has rank 1 and is not invertible.

Step 1: Compute $T(X)$ with a general X

Let $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Then:

$$T(X) = \begin{pmatrix} 1 & 2 \\ 4 & 8 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a + 2c & b + 2d \\ 4a + 8c & 4b + 8d \end{pmatrix} = \begin{pmatrix} a + 2c & b + 2d \\ 4(a + 2c) & 4(b + 2d) \end{pmatrix}.$$

Observation: Row 2 of $T(X)$ is always 4 times row 1.

(a) Kernel of T

$T(X) = 0 \iff a + 2c = 0$ and $b + 2d = 0$ (the row-2 equations are automatically satisfied).

From the two equations: $a = -2c$, $b = -2d$. Free variables: c, d .

$$X = \begin{pmatrix} -2c & -2d \\ c & d \end{pmatrix} = c \underbrace{\begin{pmatrix} -2 & 0 \\ 1 & 0 \end{pmatrix}}_{B_1} + d \underbrace{\begin{pmatrix} 0 & -2 \\ 0 & 1 \end{pmatrix}}_{B_2}.$$

$$\text{Basis of Ker}(T): \left\{ \begin{pmatrix} -2 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -2 \\ 0 & 1 \end{pmatrix} \right\}. \quad \dim \text{Ker}(T) = 2.$$

(b) Image of T

Every matrix in $\text{Im}(T)$ has the form $\begin{pmatrix} \alpha & \beta \\ 4\alpha & 4\beta \end{pmatrix}$ with $\alpha = a + 2c$, $\beta = b + 2d$. Since a, c (resp. b, d) are arbitrary, α (resp. β) can take any real value.

$$\begin{pmatrix} \alpha & \beta \\ 4\alpha & 4\beta \end{pmatrix} = \alpha \underbrace{\begin{pmatrix} 1 & 0 \\ 4 & 0 \end{pmatrix}}_{C_1} + \beta \underbrace{\begin{pmatrix} 0 & 1 \\ 0 & 4 \end{pmatrix}}_{C_2}.$$

$$\text{Basis of Im}(T): \left\{ \begin{pmatrix} 1 & 0 \\ 4 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 4 \end{pmatrix} \right\}. \quad \dim \text{Im}(T) = 2.$$

Check Rank-Nullity Theorem: $2 + 2 = 4 = \dim \mathbb{R}^{2 \times 2}$. ✓

Exercise 38

Problem. Does there exist a linear transformation with the given properties? If yes: example. If no: proof.

- (a) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with $\text{Ker}(T) = \text{Im}(T)$.
- (b) $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ with $\text{Ker}(T) = \text{Im}(T)$.
- (c) $T: \mathbb{R}^6 \rightarrow \mathbb{R}^3$ with $\dim \text{Ker}(T) = 2$.

What do the terms mean?

- **Key formula:** Rank-Nullity Theorem $\dim \text{Ker}(T) + \dim \text{Im}(T) = \dim V$ (domain).
- **Trick:** If $\text{Ker}(T) = \text{Im}(T)$ is required, then both spaces must in particular have **the same dimension**.
- **Trick:** $\dim \text{Im}(T)$ is always $\leq \dim W$ (codomain).

(a) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with $\text{Ker} = \text{Im}$

Dimension check. Let $d = \dim \text{Ker} = \dim \text{Im}$ (equal because the spaces are supposed to be equal). Rank-Nullity Theorem:

$$d + d = 2 \Rightarrow d = 1.$$

Both spaces are 1-dimensional. This is consistent – such a T *could* exist.

Construct example. Idea: T should map everything to a 1-dim subspace W ($\rightarrow \text{Im} = W$), and W should be exactly the kernel.

Attempt: $T(a, b) = (b, 0)$.

- $\text{Ker}(T)$: $T(a, b) = (0, 0) \iff b = 0$. So $\text{Ker}(T) = \{(a, 0) : a \in \mathbb{R}\} = \text{span}\{(1, 0)\}$.
- $\text{Im}(T)$: $\{(b, 0) : b \in \mathbb{R}\} = \text{span}\{(1, 0)\}$.
- $\text{Ker}(T) = \text{Im}(T)$ ✓

Yes, e.g. $T(a, b) = (b, 0)$ satisfies $\text{Ker}(T) = \text{Im}(T) = \text{span}\{(1, 0)\}$.

(b) $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ with $\text{Ker} = \text{Im}$

Dimension check. Again let $d = \dim \text{Ker} = \dim \text{Im}$. Rank-Nullity Theorem:

$$d + d = 3 \Rightarrow 2d = 3 \Rightarrow d = \frac{3}{2}.$$

d is the dimension of a vector space – must be an **integer**. $\frac{3}{2}$ is not. **Contradiction.**

No, no such T exists.

Reason: from $\text{Ker} = \text{Im}$ follows $2d = 3$, which is unsolvable for integer d .

Remark: The argument shows in general: $\text{Ker}(T) = \text{Im}(T)$ is only possible for *even* dimensions of the domain.

(c) $T: \mathbb{R}^6 \rightarrow \mathbb{R}^3$ **with** $\dim \mathbf{Ker}(T) = 2$

Dimension check. From the Rank-Nullity Theorem:

$$\dim \mathbf{Ker}(T) + \dim \mathbf{Im}(T) = \dim \mathbb{R}^6 = 6 \Rightarrow \dim \mathbf{Im}(T) = 6 - 2 = 4.$$

But $\mathbf{Im}(T) \subseteq \mathbb{R}^3$, so $\dim \mathbf{Im}(T) \leq 3$. However, we'd have $\dim \mathbf{Im}(T) = 4 > 3$.
Contradiction.

No, no such T exists.

Reason: from $\dim \mathbf{Ker} = 2$ follows $\dim \mathbf{Im} = 4$, but $\dim \mathbf{Im} \leq \dim \mathbb{R}^3 = 3$.